



Armada Kalamunda Group

Thrombolysis Guideline

1. Purpose

Guideline for thrombolytic treatment of Acute ST Elevation Myocardial Infarction (STEMI), where timely access to Percutaneous Coronary Intervention (PCI) is delayed/unable to be performed, and for high risk Pulmonary Embolism (PE).

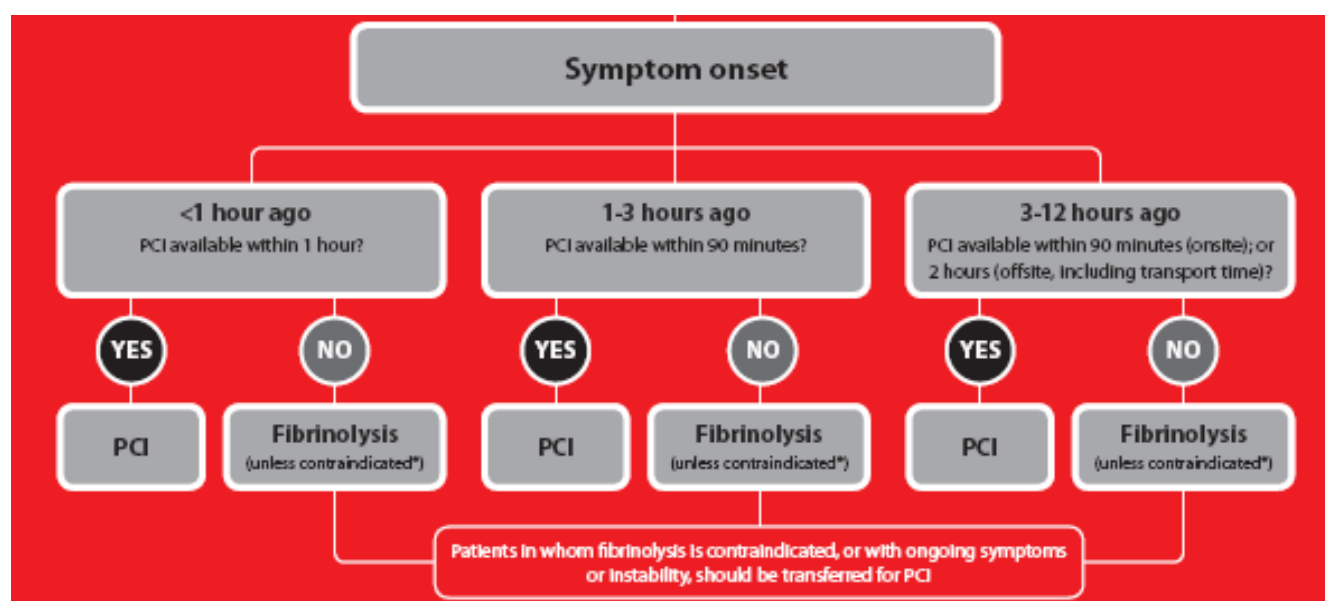
2. Indications

2.1 Acute STEMI

Refer to [AKG STEMI Guideline](#)

- Symptoms consistent with Acute Coronary Syndrome; AND ECG Changes:
 - Persistent ST elevation of >1 mm in 2 contiguous leads; or
 - New left bundle branch block
- Duration of pain less than 12 hours
- Patient not a candidate for primary angioplasty (D/W Cardiology at RPH)
- Maximum benefit is achieved when thrombolysis is given within 30 minutes of arrival to the emergency department (ED)

Thrombolysis only to be administered after discussion with **Cardiology** team at RPH



Before referencing this guideline document please ensure that you have the latest version from the [Policy Library](#) information hub page.

2.2 High Risk Pulmonary Embolus

Refer to "[High Risk Pulmonary Embolism](#)" Guideline from either FSH or RPH

- High Risk Pulmonary Embolus defined as clinical or radiological diagnosis of pulmonary embolism:
 - requiring cardiopulmonary resuscitation, or
 - with haemodynamic instability
- Haemodynamic instability defined as:
 - Systolic blood pressure <90 mmHg for ≥ 15 minutes with signs of end organ hypoperfusion, if not caused by new-onset arrhythmia, hypovolaemia or sepsis
 - or a systolic blood pressure drop of ≥ 40 mmHg ≥15 min with signs of end organ hypoperfusion, if not caused by new-onset arrhythmia, hypovolaemia or sepsis
 - Need for vasopressor support to maintain adequate organ perfusion and a systolic blood pressure >90 mmHg

Thrombolysis only to be administered after risk stratification and discussion with
Respiratory Medicine team at RPH
Likely to involve multiple subspecialty discussion – early involvement of ED
Consultant is mandatory

2.3 Cardiac Arrest where Pulmonary Embolism is the suspected aetiology

[ANZCOR guidelines](#):

- Routine administration of fibrinolytics for the treatment of in-hospital and out-of-hospital cardiac arrest is not recommended [Class A; Expert consensus opinion]
- Fibrinolysis should be considered in adult patients with cardiac arrest with proven or suspected pulmonary embolism (CoSTR 2015, weak recommendation, very low-quality evidence).
- If a fibrinolytic drug is given in these circumstances, consider performing CPR for at least 60–90 min before termination of resuscitation attempts [Class A; Expert consensus opinion].

3. Thrombolytic agent administration

TENECTEPLASE is the preferred agent for thrombolysis of both STEMI and high-risk pulmonary embolism

Refer to Tenecteplase Specialised Drug Guideline on [FSFHG Pharmacy Department Medication Administration Guidelines](#) for use of agent

Pre-administration:

- Consider absolute and relative contraindications listed and seek expert advice if required
- Obtain informed consent (if possible) and document in medical record
- Attached to cardiac monitor, obtain 12-lead ECG and record vital signs
- Establish IV access x2
- Collect blood for FBC, U&E, Troponin, Coags and G+H
- Apply pressure dressings to puncture sites and superficial wounds
- Consider IV GTN infusion for SBP >180mmHg or DBP >110mmHg

- Therapeutic anticoagulation with IV heparin should be started concomitantly as per AKMR173 Anticoagulation Medication Chart
- Note that, for STEMI thrombolysis, patient should be preloaded with Aspirin 300mg PO and Clopidogrel 300mg PO. If Ticagrelor has already been given to patient, as per [AKG STEMI Guideline](#), do not need to load with Clopidogrel.

4. Disposition

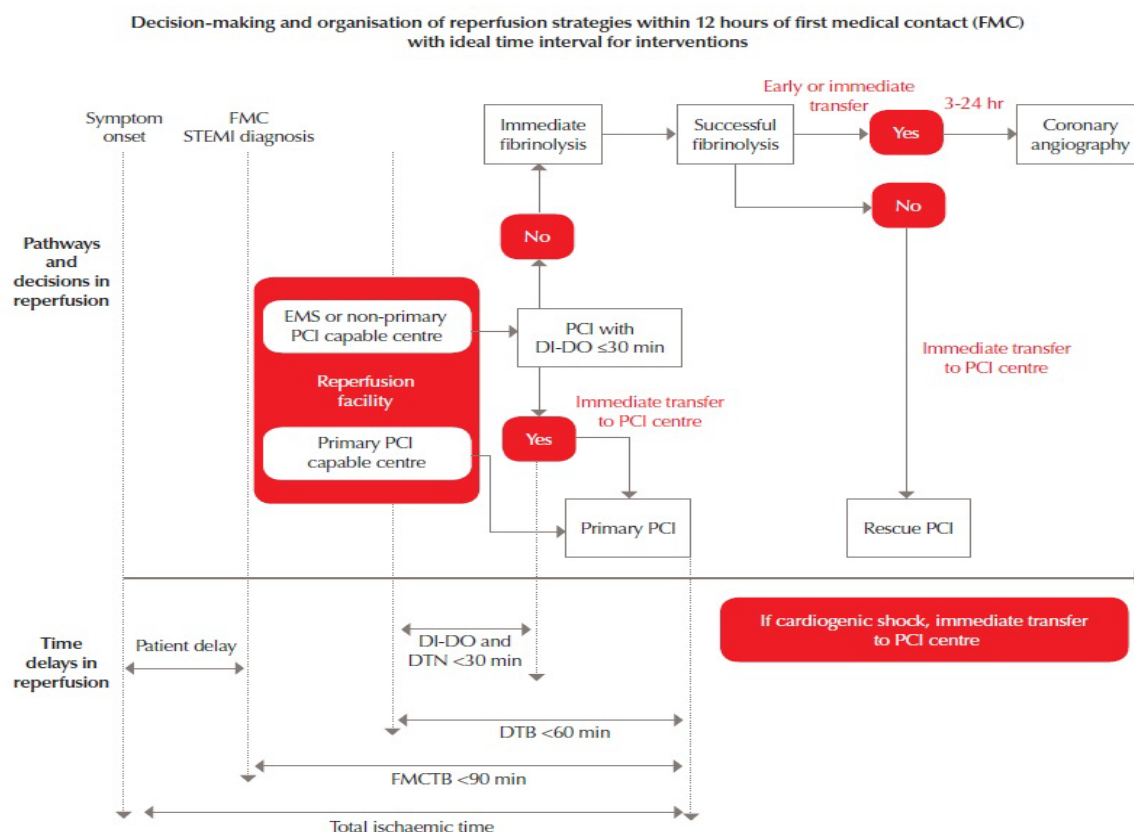
Transfer under Cardiology (CCU) or Respiratory team (HDU/ICU admission) will usually be required ASAP post thrombolysis. Contact the appropriate team. Medical escort will be required in most cases

4.1 Rescue angioplasty for failed reperfusion post thrombolysis for STEMI

Features suggestive of failed reperfusion include:

- no relief of symptoms
- persistent haemodynamic or electrical instability
- <50% reduction in initial ST-segment elevation within 60-90 minutes of therapy

These patients require **Urgent transfer** to a tertiary centre for PCI



5. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095/18](#)
- HDWA [Consent to Treatment Policy MP 0175/22](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Match EMHS: 21](#)

6. Supporting information

The following documents support the implementation of this policy:

- [ST-Elevation Myocardial Infarction \(STEMI\) / Primary Angioplasty Guideline EMD-GUI-0426-22](#)
- [Medication Management Procedure AKG-PRO-0255-22](#)

7. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated with time to reperfusion, ATS compliance.

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.01
 - Standard 6 – Action 6.04 O
 - Standard 6 – Action 6.07
 - Standard 6 – Action 6.09
 - Standard 6 – Action 6.11 References
1. **ANZCOR Guidelines**, Acute Coronary Syndromes, Guideline 14.3 – Acute Coronary Syndromes: Reperfusion Strategy – access [here](#)
 2. **ANZCOR Guidelines**, Adult Advanced Life Support, Guideline 11.5 – Medications in Adult Cardiac Arrest, 4.4 Thrombolytics – access [here](#)
 3. National Heart Foundation of Australia & Cardiac Society of Australia and New Zealand: Australian Clinical Guidelines for the Management of Acute Coronary Syndromes 2016 – access [here](#)
 4. Jaff MR, et al; Management of massive and submassive pulmonary embolism, iliofemoral deep vein thrombosis, and chronic thromboembolic pulmonary hypertension: a scientific statement from the American Heart Association. Circulation. 2011 Apr 26;123(16):1788-830
 5. Sharifi M, Bay C, Skrocki L, Rahimi F, Mehdipour M; “MOPETT” Investigators. Moderate pulmonary embolism treated with thrombolysis (from the “MOPETT” Trial). Am J Cardiol. 2013 Jan 15;111(2):273-7

9. Document owner

Consultant Emergency Physician

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10. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V5	06/02/2024	06/02/2027	Scheduled review

11. Authorisation

This guideline has been authorised and issued by Director Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
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